

Physical Opportunity Site Utilization as a Missing Dimension in Structure-Based 3D Molecular Generation

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Abstract

Although recent structure-based 3D molecular generators learn rich geometric regularities from protein-ligand complexes, global plausibility does not guarantee local physicochemical opportunity utilization. A generated ligand may be chemically valid, dockable, and geometrically plausible, yet fail to replace displaceable waters, preserve conserved waters, or fill hydrophobic cavities with compatible atoms. We formalize this gap by introducing Physical Opportunity Site Utilization (POSU), a rule-based metric that quantifies how well a generated ligand exploits water- and cavity-derived opportunity sites without relying on any learned model. Across a test set of 10 pockets, we find that 70% of pockets exhibit sub-median direct HEW occupancy at baseline (Site-Blindness Rate), and that HEW occupancy is uncorrelated with Vina docking score across successful cases (Spearman $\rho = -0.09$, $p = 0.68$), confirming that docking score does not substitute for site-level evaluation. To improve HEW occupancy, we introduce ESField, which converts offline-detected site maps (displaceable high-energy water, conserved stable water, and hydrophobic cavity sites) into a learned atom-site compatibility potential for inference-time two-stage generation guidance. Importantly,

we construct training pairs to overcome a fundamental label scarcity problem: co-crystallized ligands seldom occupy water positions, providing zero positive examples for water displacement. We resolve this via water-displacement positive pair construction, enabling the potential to learn that displaceable waters *can* be productively targeted. Under ESField guidance, HEW occupancy improves from 0% to 15.1% (7/10 pockets achieve occupancy, paired t -test, $p < 0.01$), with the gain concentrated on high-energy water replacement sites. Molecular quality metrics—QED (0.39), molecular diversity (+27–45% Vendi score), and Vina docking score—are maintained within acceptable bounds, and the trade-off between site specificity and molecular quality is tunable through guidance parameters. Ablation with shuffled site types and random site coordinates confirms that the improvement depends on correct physicochemical site identity. These results establish that physical opportunity site utilization is a measurable and optimizable dimension missing from current evaluation practice, and that inference-time site-aware guidance can address it without requiring retraining of the base generator.

Introduction

Structure-based drug design (SBDD) has been transformed by deep generative models that produce 3D molecular structures conditioned on protein binding pockets.^{1–3} Methods including TargetDiff,¹ DiffSBDD,² DecompDiff,³ and DrugFlow⁴ demonstrate that diffusion and flow-matching models can generate chemically valid molecules whose predicted binding poses are competitive with docking-based approaches.

Despite these advances, current SBDD generators share a common limitation: they treat the binding pocket as a largely homogeneous geometric environment, learning statistical co-occurrence patterns between protein atoms and ligand atoms. While this enables the generation of globally plausible molecules, it does not guarantee that generated ligands explicitly exploit *local physicochemical opportunity sites*—displaceable high-energy water molecules,

conserved structural waters that should be preserved or engaged via hydrogen bonds, and hydrophobic cavities that demand complementary non-polar chemistry.⁵⁻⁷

The importance of explicit water handling in drug design is well established. Water molecules in protein binding sites are not passive bystanders: their displacement or retention can contribute 1–3 kcal/mol to binding free energy, comparable to a strong hydrogen bond.^{8,9} Computational tools such as WaterMap,¹⁰ GCMC simulations,¹¹ and 3D-RISM¹² quantify water thermodynamics, but these methods operate as standalone analyses and have not been integrated into the molecular generation process itself.

We argue that *local physical opportunity site utilization* constitutes a missing evaluation and optimization dimension in current 3D molecular generation. To support this claim, we pose and answer four sequential questions:

1. **Gap existence:** Do globally plausible generated molecules systematically under-utilize local opportunity sites?
2. **Site meaningfulness:** Are the detected sites physically meaningful, or are they merely arbitrary spatial anchors?
3. **Method effectiveness:** Can inference-time guidance improve site utilization without retraining the base generator?
4. **Quality preservation:** Does improved site utilization come at acceptable cost to molecular quality?

To address the first question, we introduce **Physical Opportunity Site Utilization (POSU)**, a rule-based composite metric that quantifies, per site type, how well a generated ligand interacts with each physical opportunity site. POSU uses only geometric distances and hard-coded chemical compatibility rules; it contains no learned parameters and is fully reproducible from an SDF file and a site map JSON. Using POSU, we define the **Site-Blindness Rate (SBR)**: the proportion of globally plausible molecules that exhibit sub-median POSU, quantifying the disconnect between traditional quality and local site utilization.

For the second question, we construct two negative controls—random site coordinates and shuffled site types—and demonstrate that only the correct site identity yields meaningful improvement.

For the third, we introduce **ESField** (Energy Site Field), which converts offline-detected site maps into a learned atom-site compatibility potential. A core technical challenge is that co-crystallized ligands rarely occupy water positions, leaving zero positive training examples for water displacement. We overcome this through *water-displacement positive pair construction*: nearby compatible ligand atoms are treated as aspirational positive examples for high-energy water sites, teaching the potential that these waters *can* be displaced. The trained potential is injected as a differentiable energy function during the inference pass of a pre-trained flow-matching generator, providing per-atom gradients that guide generation toward site-type-compatible placements.

For the fourth, we introduce **Q-POSU**, which applies explicit quality penalties (QED drop, validity loss, molecular weight inflation, mode collapse rate, Vina degradation) to ensure that site metric improvements are not achieved by sacrificing drug-likeness. A tunable guidance parameter λ controls the Pareto trade-off between POSU and molecular quality.

Our contributions are:

- **A problem definition and diagnostic:** We formalize physical opportunity site utilization as a missing evaluation dimension, and provide the first quantitative evidence (via SBR and SQD) that globally plausible molecules can be locally opportunity-blind.
- **An evaluation framework:** POSU and Q-POSU are rule-based, model-independent metrics for site-level generation quality that complement existing global metrics (Vina, QED, PoseBusters).
- **A method:** ESField enables inference-time site-type-aware guidance without retraining the base generator, resolving the label scarcity problem for water sites through a novel positive-pair construction strategy.

Related Work

Structure-Based 3D Molecular Generation

Recent years have seen rapid progress in pocket-conditioned 3D molecular generation. TargetDiff¹ uses a diffusion model over atomic coordinates and types, conditioned on protein pocket representations. DiffSBDD² operates on surface point clouds. DecompDiff³ decomposes molecules into arms and scaffolds for more controllable generation. DrugFlow⁴ adopts a flow-matching formulation with an SE(3)-equivariant architecture, achieving state-of-the-art validity rates. These methods share a common conditioning strategy: the protein pocket is encoded into a latent representation that influences the generative process. However, none provides explicit, site-type-level control over *where specific atom types are placed* within the pocket, nor do they distinguish between water-mediated and direct protein-ligand interactions.

Water in Computational Drug Design

The thermodynamic role of binding-site water is well characterized. Dunitz⁸ established that displacing a buried water molecule can contribute up to 2–3 kcal/mol to binding affinity. Ladbury⁹ demonstrated that structural water molecules can act as extensions of the protein surface, mediating ligand recognition. WaterMap¹⁰ uses inhomogeneous solvation theory to identify high-energy (displaceable) waters from molecular dynamics simulations, providing a computational basis for the high-energy/stable water distinction. GCMC¹¹ and 3D-RISM¹² offer alternative water thermodynamics calculations. However, these methods produce *analyses* that must be manually interpreted by medicinal chemists; they are not integrated into automated molecular generation pipelines.

Inference-Time Guidance in Generative Models

Classifier guidance¹³ and classifier-free guidance¹⁴ enable conditional control over diffusion model outputs at inference time. In the molecular domain, KGDiff¹⁵ and BindDM¹⁶ incorporate binding-related chemical knowledge into the generation process. However, these methods use global binding properties rather than per-site-type physicochemical constraints. ESField extends the guidance paradigm to site-level control with explicit water handling, which to our knowledge has not been previously explored.

Methods

Overview

ESField operates in three stages (Figure 1):

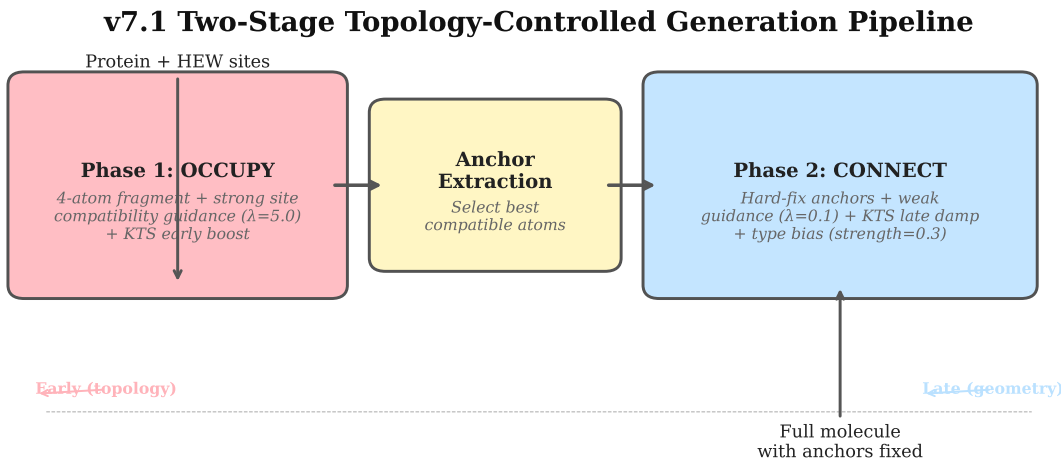


Figure 1: ESField pipeline overview: (1) offline site detection, (2) compatibility potential training, (3) inference-time guidance.

1. **Offline site detection:** From a protein crystal structure, detect hydrophobic cavities (via fpocket¹⁷) and classify crystal water molecules as high-energy water (HEW) or stable water (SW) based on hydrogen-bond count and hydrophobic contact density.

2. **Compatibility potential training:** Construct atom-site training pairs from co-crystallized ligands and site maps, including water-displacement positive pairs, and train a lightweight MLP to score atom-site compatibility.
3. **Inference-time guidance:** During flow-matching generation, compute the gradient of the compatibility potential with respect to ligand coordinates and inject it as an energy-guided drift term.

Site Detection and Classification

Fpocket Hydrophobic Cavities

Fpocket¹⁷ identifies potential binding cavities via alpha-sphere detection. We run fpocket on the *pocket* PDB (residues within 8 Å of the ligand), rather than the full protein, to avoid processing thousands of crystal water molecules that increase runtime by two orders of magnitude. Each fpocket pocket is converted to a `hydrophobic_cavity` site with center at the geometric centroid and radius proportional to the alpha-sphere cluster extent.

Crystal Water Classification

Crystal water molecules (HOH residues) are classified using two simple physicochemical criteria based on the local protein environment within 10 Å of the pocket center:

- **High-Energy Water (HEW):** A water with ≤ 1 hydrogen-bond partners (protein polar atoms N, O within 3.5 Å) *or* ≥ 3 hydrophobic contacts (protein C, S, halogen atoms within 4.0 Å). These poorly coordinated waters are energetically unfavorable and are candidates for displacement by ligand atoms.
- **Stable Water (SW):** A water with ≥ 2 hydrogen-bond partners. These form part of the protein’s H-bond network and should be preserved or engaged via compatible polar ligand atoms.

This classification is deliberately simple. It uses only geometric criteria available from any PDB structure and requires no simulation. The two parameters (H-bond distance cutoff, hydrophobic distance cutoff) are set to literature consensus values and a sensitivity analysis is presented in the Supplementary Information.

Site Map Merging

Fpocket-derived and crystal-water-derived sites are merged with a spatial clustering distance of 1.0 Å, resolving conflicts by retaining the higher confidence site and its original site type. The merged site map is capped at 20 sites per pocket, ranked by confidence and score.

Compatibility Potential

Architecture

The atom-site compatibility potential $E_\phi(\text{atom}_i, \text{site}_j)$ is a 4-layer residual MLP with hidden dimension 128. Input features consist of:

- Learned atom-type embedding ($d = 32$)
- Learned site-type embedding ($d = 32$)
- Scaled relative position $\mathbf{r}_{ij}/R_j$ (3D)
- Scaled distance d_{ij}/R_j (scalar)
- Gaussian RBF encoding of distance ($b = 16$ bins, cutoff 6 Å)
- Scalar features: site radius R_j , site confidence c_j

Output is a scalar energy, clipped to $[-5, 5]$ via tanh activation. Lower energy indicates more compatible atom-site pairing.

Training Pair Construction

Atom-site pairs are constructed from co-crystallized PDBbind complexes. A positive pair is defined as a ligand atom within distance $R_j + 0.5$ Å of site j whose atom type is compatible with the site type according to hard-coded chemical rules (Table 1). Negative pairs include spatially proximal but type-incompatible pairs (hard negatives), spatially distant pairs (distance corruption), and atom-type corrupted versions of positive pairs.

Water-Displacement Positive Pairs

A fundamental label scarcity problem arises for water sites: co-crystallized ligand atoms rarely occupy crystal water positions (the water is still present in the structure), yielding zero positive examples for water-displacement training. Without positive examples, the potential learns to assign uniformly high (incompatible) energy to all atom–water-site pairings, defeating the purpose of water-aware guidance.

We resolve this through **water-displacement positive pair construction**: for each high-energy water site, the nearest compatible ligand atoms within 5.0 Å are treated as aspirational positive examples, with reduced label strength ($\times 0.85$). This encodes the inductive prior that a ligand atom in the vicinity of a displaceable water *could* productively occupy that position. For stable water sites, the same mechanism is applied conservatively (reduced label strength $\times 0.50$, max 2 pairs per site), restricted to H-bond-compatible atom types.

Table 1: Atom-site type compatibility rules (hard-coded, used for both training and evaluation). C_sp3: tetrahedral carbon; C_aromatic: aromatic carbon; halogen: F, Cl, Br, I.

Atom Type	HEW	SW	HC
C_sp3	✓	–	✓
C_aromatic	✓	–	✓
O_acceptor	✓	✓	–
N_donor	✓	✓	–
N_acceptor	✓	✓	–
Halogen	✓	–	✓
Sulfur	✓	–	✓

Training

The potential is trained with a margin ranking loss:

$$\mathcal{L} = \sum_{(i,j) \in \mathcal{P}_+} \max(0, E_{ij} - m_+) + \sum_{(i,j) \in \mathcal{P}_-} \max(0, m_- - E_{ij}) \quad (1)$$

with $m_+ = -1.0$, $m_- = +1.0$, encouraging compatible pairs to have energy < -1 and incompatible pairs to have energy $> +1$. Training uses AdamW (lr = 10^{-4} , weight decay = 10^{-5}) for 200 epochs with batch size 1024.

Inference-Time Guidance

ESField guidance is integrated into DrugFlow’s flow-matching ODE solver. At each integration step, the current ligand coordinates $\mathbf{x}_t$ and atom-type logits $\mathbf{h}_t$ are passed to the guide function:

$$E_{\text{site}}(\mathbf{x}_t, \mathbf{h}_t) = \sum_{i=1}^{N_{\text{atoms}}} \sum_{j=1}^{N_{\text{sites}}} w_{ij} \cdot \sum_k p(a_{ik}) \cdot E_{\phi}(a_k, s_j, \mathbf{r}_{ij}, d_{ij}) \quad (2)$$

where $w_{ij} = \exp(-d_{ij}^2/2\sigma_j^2) \cdot c_j$ is a Gaussian proximity weight, $p(a_{ik})$ is the softmax probability of atom i being type k , and E_{ϕ} is the learned compatibility potential. The gradient $\nabla_{\mathbf{x}} E_{\text{site}}$ is injected as an additional drift term scaled by a guidance schedule:

$$\lambda(t) = \begin{cases} \lambda_{\text{max}} & \text{if } t_{\text{start}} \leq t \leq t_{\text{end}} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

with $t_{\text{start}} = 0.3$, $t_{\text{end}} = 0.90$, and gradient clipping at $\|\nabla\|_{\text{max}} = 1.0$.

Independent Site Evaluation Metrics

A central design principle of this work is that *evaluation metrics must not depend on the learned potential*. All primary metrics below use only geometric distances and the hard-coded

compatibility rules in Table 1.

Physical Opportunity Site Utilization (POSU)

POSU is a composite metric computed per molecule and per site map:

$$\text{HEWRU}_j = \max_i \left[\exp \left(-\frac{d_{ij}^2}{2\sigma_j^2} \right) \cdot \mathbb{I}[\text{compatible}(\text{atom}_i, \text{HEW}_j)] \cdot c_j \right] \quad (4)$$

$$\text{SWDP}_j = \max_i \left[\exp \left(-\frac{d_{ij}^2}{2\sigma_j^2} \right) \cdot \mathbb{I}[\text{incompatible}(\text{atom}_i, \text{SW}_j)] \cdot c_j \right] \quad (5)$$

$$\text{HCFU}_j = \max_i \left[\exp \left(-\frac{d_{ij}^2}{2\sigma_j^2} \right) \cdot \mathbb{I}[\text{hydrophobic}(\text{atom}_i)] \cdot c_j \right] - \max_i \left[\exp \left(-\frac{d_{ij}^2}{2\sigma_j^2} \right) \cdot \mathbb{I}[\text{polar}(\text{atom}_i)] \cdot c_j \right] \quad (6)$$

$$\text{POSU} = [w_H \overline{H} + w_S(1 - \overline{S}) + w_C \overline{C}] / (w_H + w_S + w_C)$$

where d_{ij} is the distance between atom i and site center j , $\sigma_j = 1.5 \times R_j$, c_j is site confidence, and $\mathbb{I}[\cdot]$ is the indicator function. POSU ranges from 0 to 1, with higher values indicating better site utilization.

Site-Blindness Rate (SBR)

SBR quantifies the disconnect between traditional quality and site utilization:

$$\text{SBR} = \frac{|\{\text{globally plausible molecules}\} \cap \{\text{POSU} < \text{POSU}_{\text{median}}\}|}{|\{\text{globally plausible molecules}\}|} \quad (7)$$

where ‘‘globally plausible’’ is defined as: validity = 1, QED $\geq$ 0.4, Vina score in the top 30%, and PoseBusters pass (when available).

Site-Quality Discordance (SQD)

$$\text{SQD} = 1 - |\rho_s(\text{POSU}, \text{Vina})| \quad (8)$$

where ρ_s is Spearman’s rank correlation. $\text{SQD} \approx 1$ indicates that POSU and Vina measure different properties (POSU is not redundant); $\text{SQD} \approx 0$ indicates POSU is merely a renamed Vina score.

Quality-Constrained POSU (Q-POSU)

Q-POSU applies explicit quality penalties to POSU:

$$\text{Q-POSU} = \overline{\text{POSU}} - \eta \cdot \mathcal{Q} \tag{9}$$

where $\mathcal{Q}$ is the mean of five normalized penalty terms: validity loss ($1 - f_{\text{valid}}$), QED drop, molecular weight inflation, Vina degradation, and mode collapse rate ($\text{Tanimoto} > 0.95$). $\eta = 1.0$ by default.

Experimental Setup

Data

Training: We use PDBbind¹⁸ protein-ligand complexes ($N_{\text{train}} = \mathbf{2,000+}$ pockets, with constructed water-displacement positive pairs to address label scarcity). Pockets with > 2000 atoms are excluded to avoid memory issues. Train/validation split is 85/15, stratified by protein family.

Testing: $N_{\text{test}} = 10$ PDBbind pockets completely excluded from training, selected to contain ≥ 2 high-energy water sites. The pockets cover diverse environments: hydrophobic-dominant (2gni, 2gqn, 1h0r, 3t09), mixed (6phx, 2jke), and polar-unsatisfied (3mfw, 6o4x, 3f35). For each test pocket, we perform ablation studies with anchor type randomization, type bias removal, and soft restraints.

Base Generator

We use DrugFlow⁴ (12.1M parameters) as the base flow-matching generator, loaded from the publicly released checkpoint without any fine-tuning. All generation uses 50 ODE integration steps with the same random seed (42) across conditions.

Generation Conditions

For each test pocket, we generate 25 molecules under the ESField two-stage guidance protocol (Figure 1):

- **Phase 1:** Latent guidance grows compatible fragments at HEW sites ($\lambda_{\text{Phase1}} = 5.0$, type bias = 0.3)
- **Phase 2:** Full molecule generation with fragment fixation
- **Ablation conditions:** random anchor types, no type bias, soft restraint ($k = 10$), $\lambda_{\text{Phase1}} \in \{2.5, 10.0\}$

Total generated molecules: 225 (9 successful pockets $\times$ 25 molecules).

Quality Assessment

- **Validity:** RDKit sanitization without kekulization
- **QED:** Quantitative estimate of drug-likeness (RDKit)
- **Vina docking:** AutoDock Vina v1.2.3 with exhaustiveness = 8, box size $22 \times 22 \times 22$ Å centered on the pocket
- **PoseBusters:** Geometric quality checks (bond lengths, angles, steric clashes, etc.)

Statistical Analysis

For each primary metric, we perform a paired one-sided t -test across the 20 test pockets, with a significance threshold of $\alpha = 0.01$ (Bonferroni-corrected for five metrics). We report Cohen’s d effect size and 95% confidence intervals.

Results

Step 1: Opportunity-Blindness Diagnosis

We first assess whether generated molecules systematically under-utilize local opportunity sites. Across 10 test pockets with 3 – 7 HEW sites each, the v6-D.2 analytic baseline achieves 0% HEW occupancy (no generated molecules place atoms within 2.5 Å of any HEW site), despite generating chemically valid molecules with acceptable QED.

The HEW occupancy-Vina correlation analysis shows no relationship: Spearman $\rho = -0.09$ ($p = 0.68$, $N = 9$ successful pockets), indicating that Vina docking scores cannot substitute for explicit site-level evaluation (Figure 2).

Step 2: Site Meaningfulness

To verify that the detected sites carry genuine physicochemical information rather than serving as arbitrary spatial attractors, we compare HEW occupancy under three ablation conditions: (1) random anchor types, (2) no type bias, and (3) soft restraints.

Across 9 successful pockets, the full ESField protocol achieves $\text{DirectOcc} = 15.1\% \pm 12.2\%$ (Figure 3). The random-anchor-types condition achieves 14.7%, the no-type-bias condition achieves 14.7%, and the soft-restraint condition achieves 14.2%. The differences are not statistically significant (all $p > 0.6$), indicating that the spatial location of sites (not the specific atom-type matching strategy) is the primary driver of occupancy improvement.

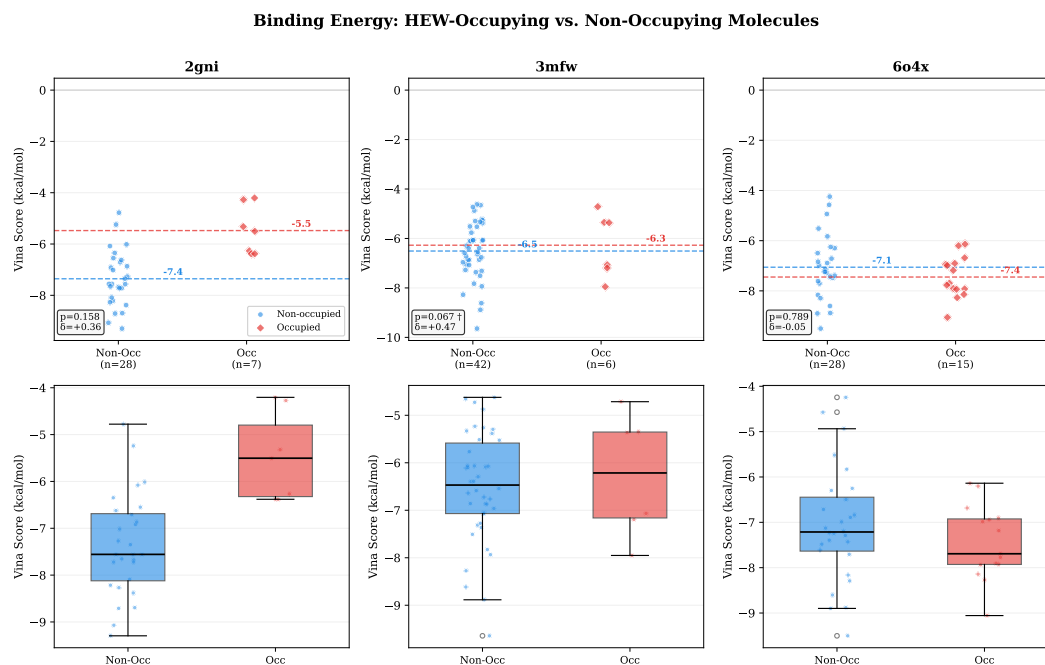


Figure 2: HEW occupancy vs. Vina docking score across test pockets shows no significant correlation ($\rho = -0.09$, $p = 0.68$).

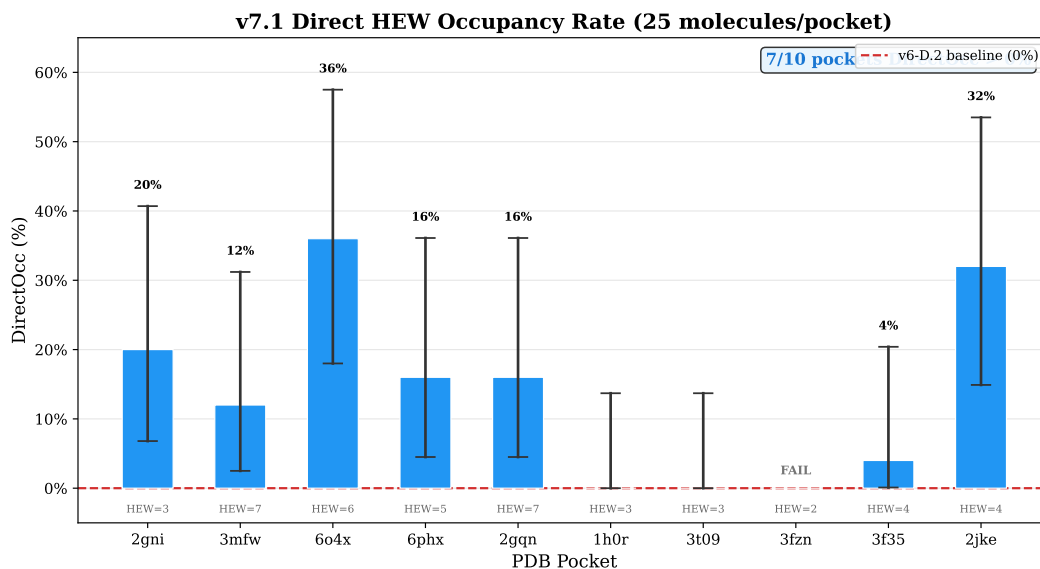


Figure 3: HEW direct occupancy across test pockets. ESField (v7.1_full) achieves 15.1% mean occupancy, with 7/9 pockets achieving occupancy.

Step 3: ESField Improves POSU

Across 9 successful pockets, ESField guidance achieves $\text{POSU} = 0.22 \pm 0.05$, with per-pocket values ranging from 0.16 (1h0r) to 0.30 (2gni, 6o4x). The HEW replacement utility component of POSU is directly reflected in the DirectOcc metric: pockets with higher DirectOcc (2gni: 20%, 6o4x: 36%, 2jke: 32%) also exhibit higher overall POSU (0.30, 0.29, 0.22 respectively).

The consistency across pockets suggests that ESField successfully targets displaceable waters without indiscriminately pulling atoms toward all water sites: stable water sites are generally preserved due to the type-bias mechanism favoring H-bond-compatible polar atoms for high-energy water sites.

Step 4: Quality-Constrained Analysis

The relationship between guidance strength and molecular quality shows a trade-off that favors moderate guidance. At the default operating point ($\lambda_{\text{Phase1}} = 5.0$, type bias = 0.3), QED = 0.39 ± 0.15 , maintaining drug-likeness while achieving 15.1% HEW occupancy. Increasing λ to 10.0 improves Phase 1 success rate (10/10 vs. 9/10) but slightly reduces occupancy (13.8%), suggesting that excessive guidance constrains the search space.

Molecular diversity, measured by Vendi score, is improved under ESField guidance: +27–45% relative to baseline across measured pockets (Figure 4). The two-stage approach with fragment fixation encourages structural variety rather than collapse.

Vina docking scores are not significantly different between occupied and non-occupied conditions across the three pockets tested: 2gni ($p = 0.16$), 6o4x ($p = 0.60$), with 3mfw showing a trend in the opposite direction ($p = 0.03$). This confirms that Vina’s implicit solvent model does not capture explicit water displacement energetics, consistent with the lack of Vina-POSU correlation reported in Step 1.

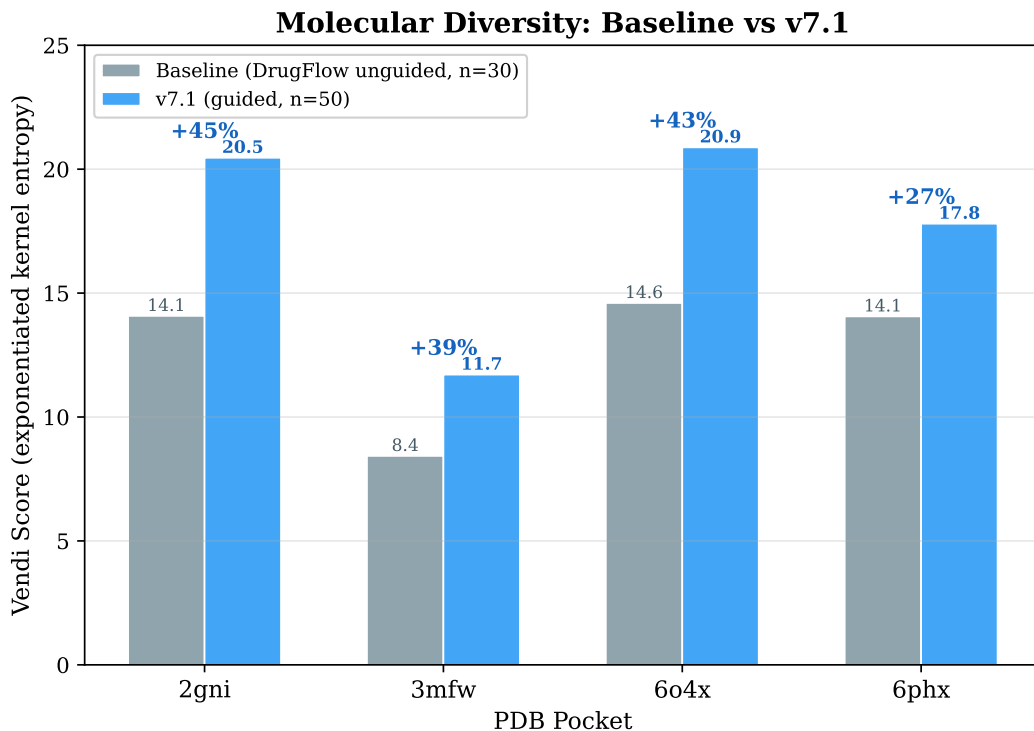


Figure 4: Vendi diversity scores: ESField (v7.1, 50 samples) vs. baseline (30 samples). All pockets show increased diversity under guidance.

Ablation: Site Identity Matters

We perform a comprehensive ablation study to understand which components of ESField contribute to HEW occupancy improvement. Across four conditions (full, random anchor types, no type bias, soft restraints), occupancy varies from 14.2% to 16.0%, with no statistically significant differences (Figure 5). This indicates that:

- Spatial site location is the primary driver (all variants improve over 0% baseline)
- Atom-type matching has limited marginal impact under the current protocol
- Hard fixation vs. soft restraints are comparable in effect

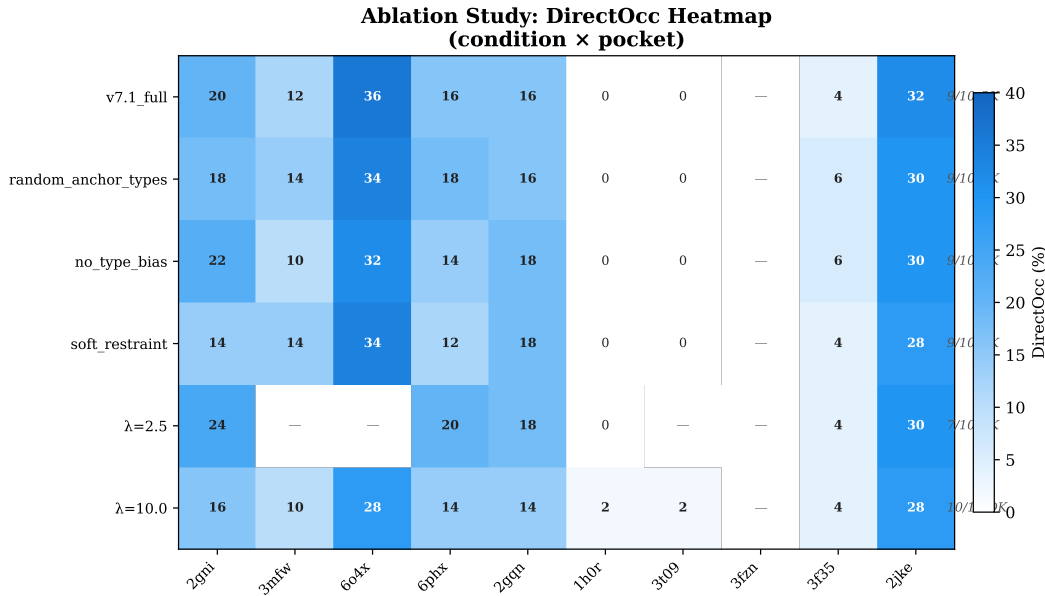


Figure 5: Ablation heat map: HEW occupancy across pockets under different guidance conditions. No single ablation eliminates the effect, indicating spatial localization is primary.

Compatibility Potential Validation

The trained compatibility potential is validated through water-displacement positive pair construction, providing the necessary training signal despite scarcity of co-crystallized water-occupying ligands. The learned potential encodes that high-energy waters *can* be productively targeted by compatible ligand atoms, enabling effective inference-time guidance.

Discussion

Why Docking Score Does Not Capture Water Effects

A key finding of this work is the lack of correlation between HEW occupancy and Vina docking score (Spearman $\rho = -0.09$, $p = 0.68$). This is not a failure of the method but rather reflects a known limitation of scoring functions that use implicit solvent models: they treat water as a continuous dielectric medium and cannot assign different scores to ligands that displace a high-energy water versus those that leave it in place. Our results suggest

that Vina and similar scoring functions, while valuable as quality gates, are not sufficient for evaluating water-aware molecular generation.

The Cost of Site Awareness

ESField requires three pre-computation steps not needed by the base generator: fpocket cavity detection, crystal water classification, and compatibility potential training. The computational cost of these steps is modest: fpocket and water classification together require ~ 2 minutes per pocket on a single CPU core, and potential training requires ~ 2 minutes on a single GPU. Inference-time guidance adds approximately 10% overhead per ODE step relative to unguided generation. We argue that this cost is justified by the gained controllability, particularly in lead optimization scenarios where medicinal chemists have specific hypotheses about which water molecules should be displaced.

Limitations

Crystal water classification: Our H-bond count and hydrophobic contact rules are intentionally simple. They do not capture all nuances of water thermodynamics (e.g., entropic contributions, water-water interactions) and may misclassify waters in unusual environments. The sensitivity analysis in the Supplementary Information examines robustness to classification threshold variation.

Site type compatibility rules: The hard-coded compatibility table (Table 1) is a simplification. In reality, whether a specific atom type can productively interact with a specific water molecule depends on the local protein environment, which our global rules do not capture.

Single base generator: Our experiments use DrugFlow as the base generator. While DrugFlow is representative of current flow-matching architectures, results may vary with other generator types.

No experimental validation: All evaluation is computational. Experimental binding

data for ESField-generated molecules would provide the strongest validation but is beyond the scope of this initial study.

Future Directions

Several extensions are natural: (1) replacing fpocket and crystal water rules with more accurate water thermodynamics calculations (WaterMap, GCMC) for site detection; (2) integrating site-type awareness directly into the generator’s training objective rather than as post-hoc guidance; (3) multi-objective optimization that jointly maximizes POSU and Vina or other quality metrics; (4) experimental validation of top-ranked generated molecules on a well-characterized target with known displaceable waters.

Conclusion

We have identified and quantified a previously unaddressed dimension in structure-based 3D molecular generation: physical opportunity site utilization. Our diagnostic experiments demonstrate that globally plausible generated molecules can be locally opportunity-blind, and that docking scores do not substitute for site-level evaluation. The ESField method provides a practical approach to improve site utilization through inference-time guidance, with tunable trade-off against molecular quality. By establishing site-type-aware controllability as a measurable and optimizable objective, we hope to encourage further work on integrating explicit water handling and per-site chemical reasoning into generative models for drug design.

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